



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



Module 2 – Regression

Dr. Nirmala Baloorkar

Assistant Professor

Department of Computer Engineering KJSSE

Supervised Learning

“

Is a type of machine learning used to train models from labeled training data.
It allows you to predict output for future or unseen data.

”

Examples of Supervised Learning

Example 1: Weather Apps

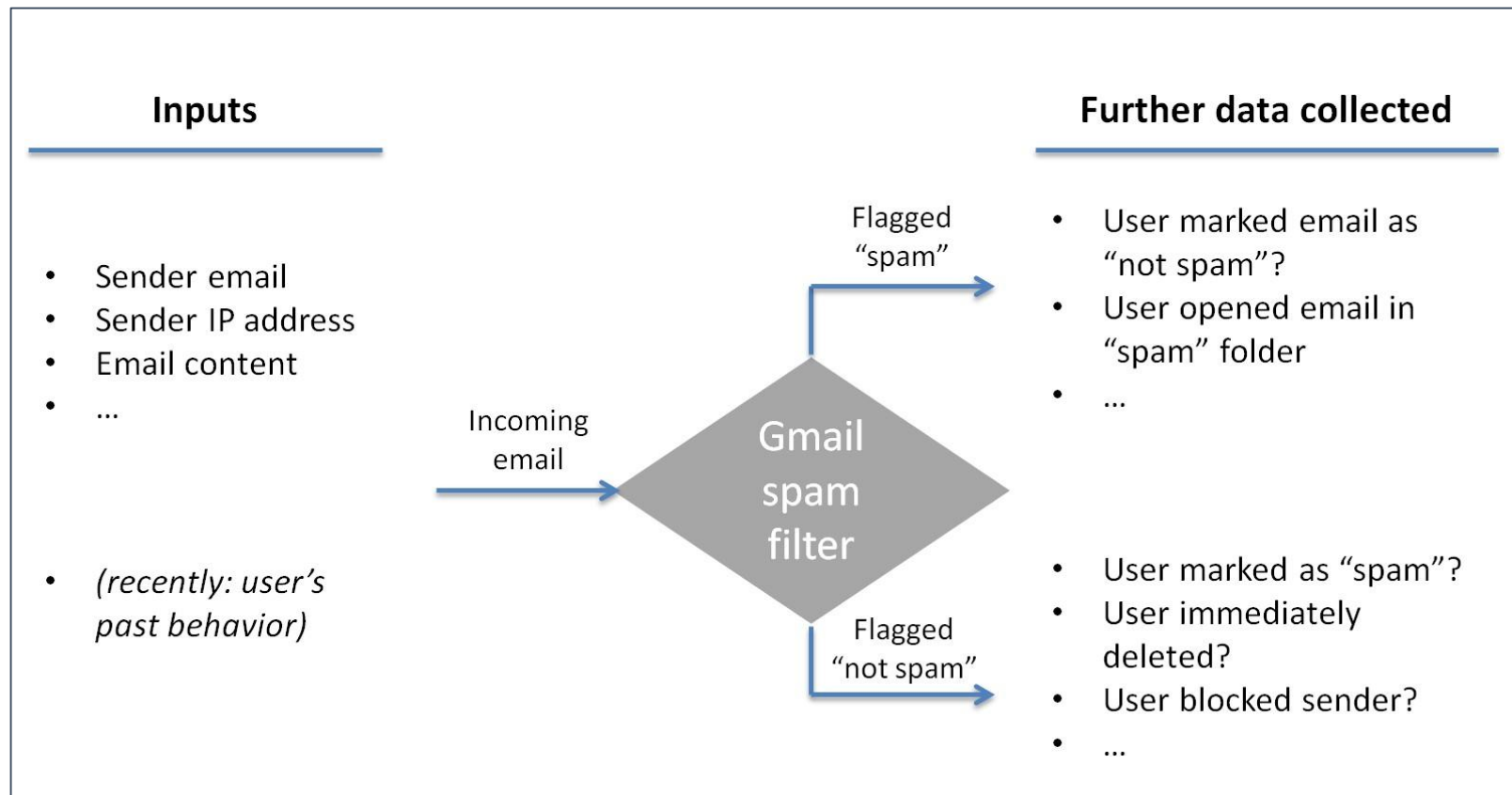
The predictions made by weather apps at a given time are based on prior knowledge and analysis of weather over a period of time for a particular place.



Examples of Supervised Learning (Contd.)

Example 2: Gmail Filters

Gmail filters, a new email into Inbox (normal) or Junk folder (Spam) based on past information of spam.



Real-Life Scenario

Ever wondered how Netflix makes recommendations?

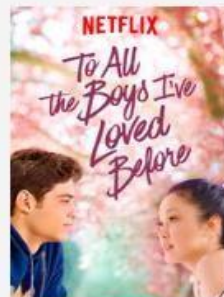
Because you added **Dark** to your List



① User, choose 3 you like

It will help us find TV programmes & films you'll love! **Click the ones you like!**

Continue



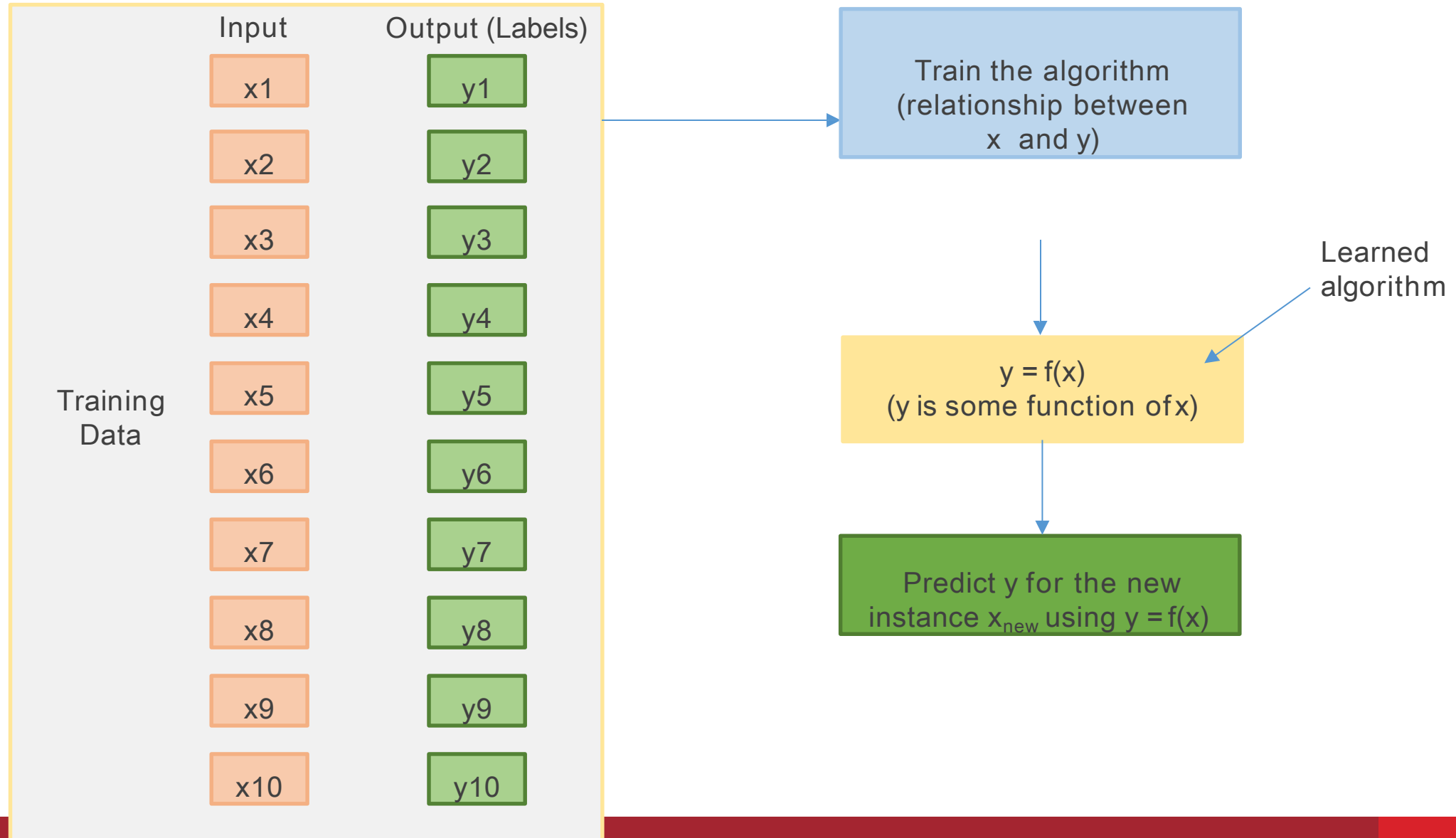
Supervised Learning: Case Study



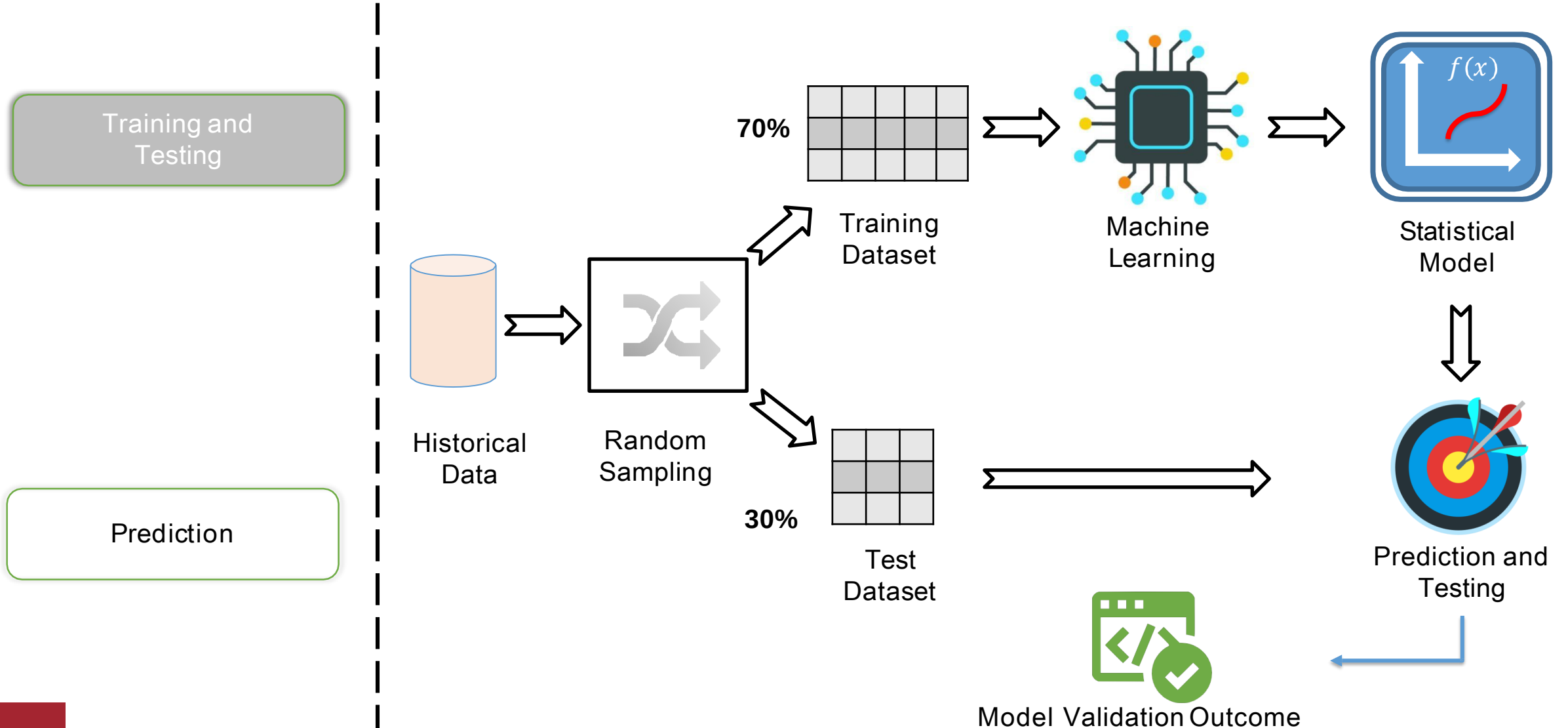
Netflix uses **supervised learning** algorithms to recommend users the shows they may watch based on the viewing history and ratings by similar classes of users



Understanding the Algorithm



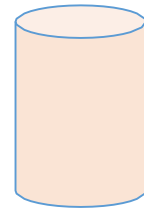
Supervised Learning Flow



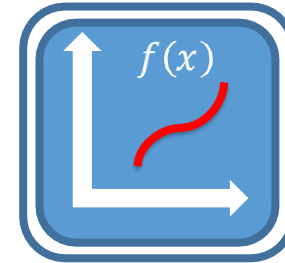
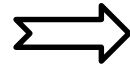
Supervised Learning Flow

Training and
Testing

Prediction



New Data



Model



Prediction
Outcome

Use the learned
algorithm $y = f(x)$ to
predict production
data

Algorithm prediction can be improved by more training data, capacity, or algorithm redesign.



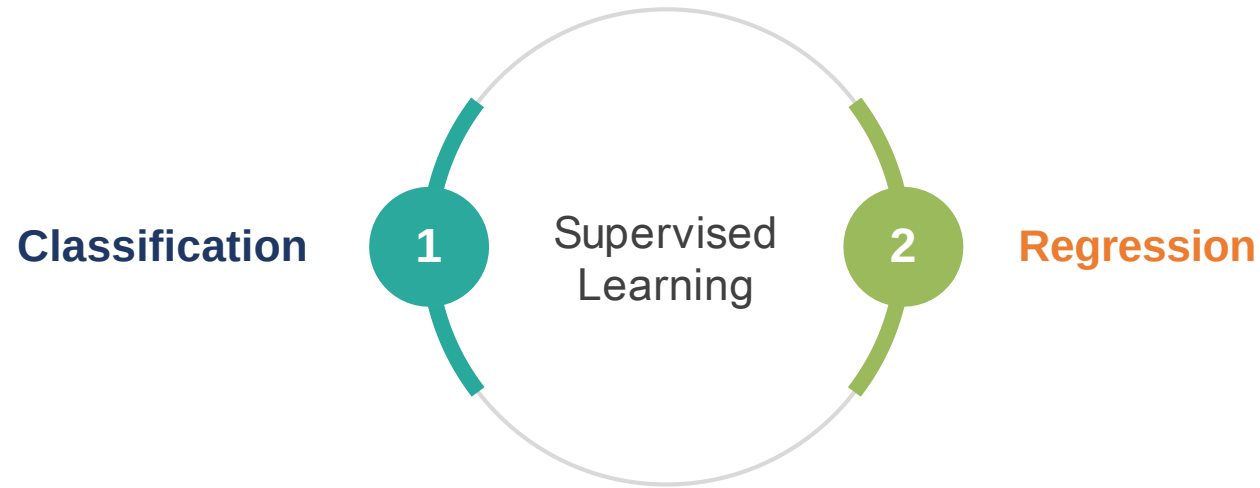
SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



Types of Supervised Learning

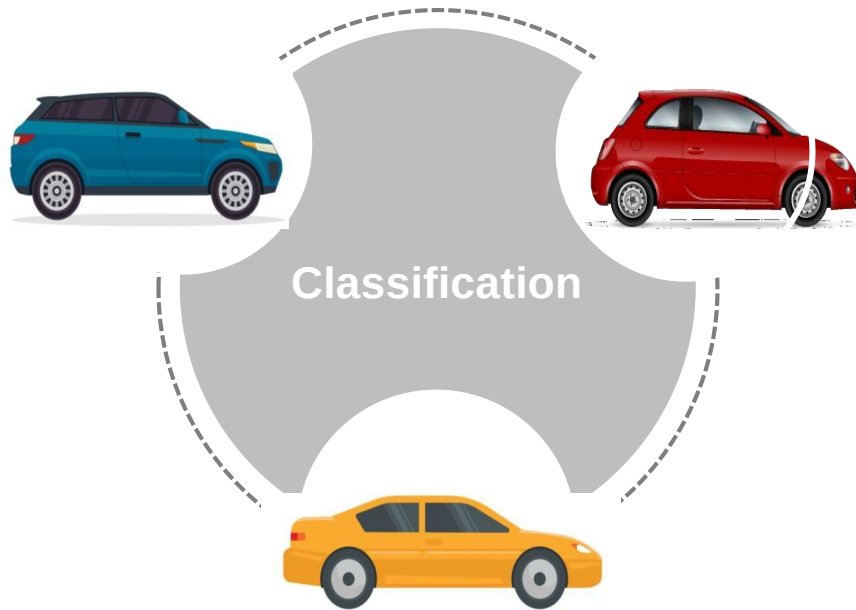
Types of Supervised Learning



In supervised learning, algorithm is selected based on target variable.

Types of Supervised Learning (Contd.)

If target variable is categorical (classes), then use classification algorithm.



In other words, classification is applied when the output has finite and discrete values.

Example: Predict the class of car given its features like horsepower, mileage, weight, colour, etc.

The classifier will build its attributes based on these features. Analysis has three potential outcomes - Sedan, SUV, or Hatchback

Types of Supervised Learning (Contd.)

If target variable is a continuous numeric variable (100–2000), then use a regression algorithm.



Example: Predict the price of a house given its sq. area, location, no of bedrooms, etc.

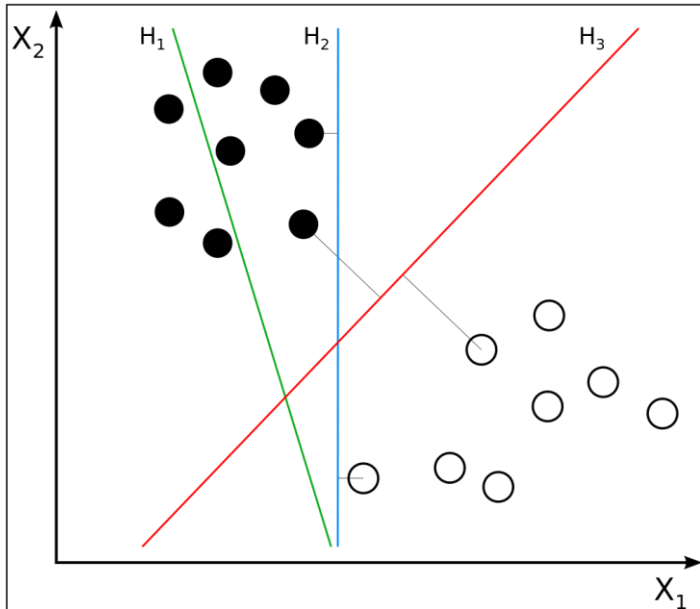
A simple regression algorithm is given below

$$y = w * x + b$$

This shows relationship between price (y) and sq. area (x) where price is a number from a defined range.

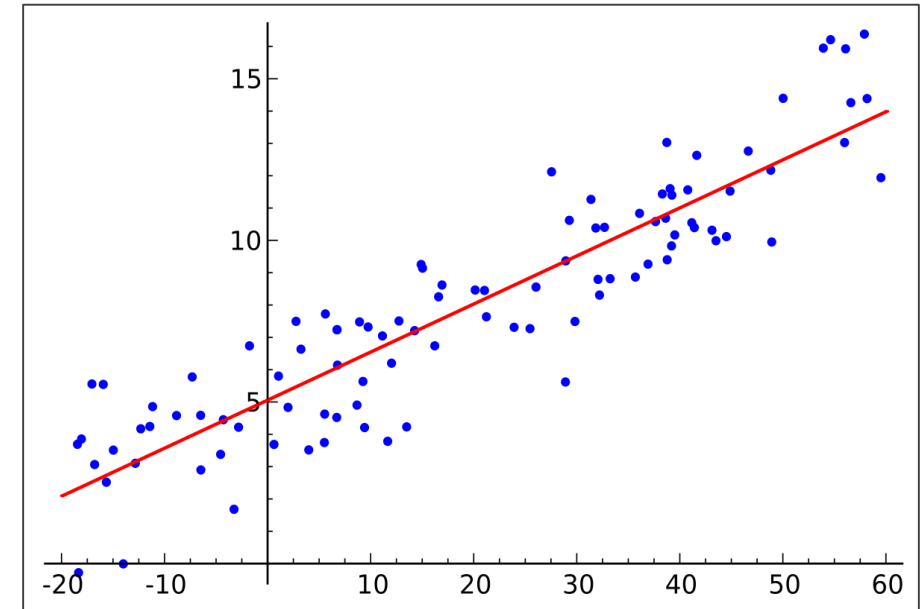
Types of Supervised Learning (Contd.)

Classification



What Class?

Regression



How much?

Regression

- Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.
- More specifically, Regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed. It predicts continuous/real values such as temperature, age, salary, price, etc.

Regression

Example: Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380
\$200	??

Now, the company wants to do the advertisement of \$200 in the year 2019 **and wants to know the prediction about the sales for this year**. So to solve such type of prediction problems in machine learning, we need regression analysis.

Regression

- Regression analysis is a statistical method that helps us to analyze and understand the relationship between two or more variables of interest.
- Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables.
- It is mainly used for prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables.

Regression

In Regression, we plot a graph between the variables which best fits the given datapoints, using this plot, the machine learning model can make predictions about the data.

Regression shows a line or curve that passes through all the datapoints on target-predictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum."

The distance between datapoints and line tells whether a model has captured a strong relationship or not.

Example1: Prediction of rain using temperature and other factors

Example2: Determining Share Market trends

Example3: Prediction of road accidents due to rash driving.

Terminologies in Regression

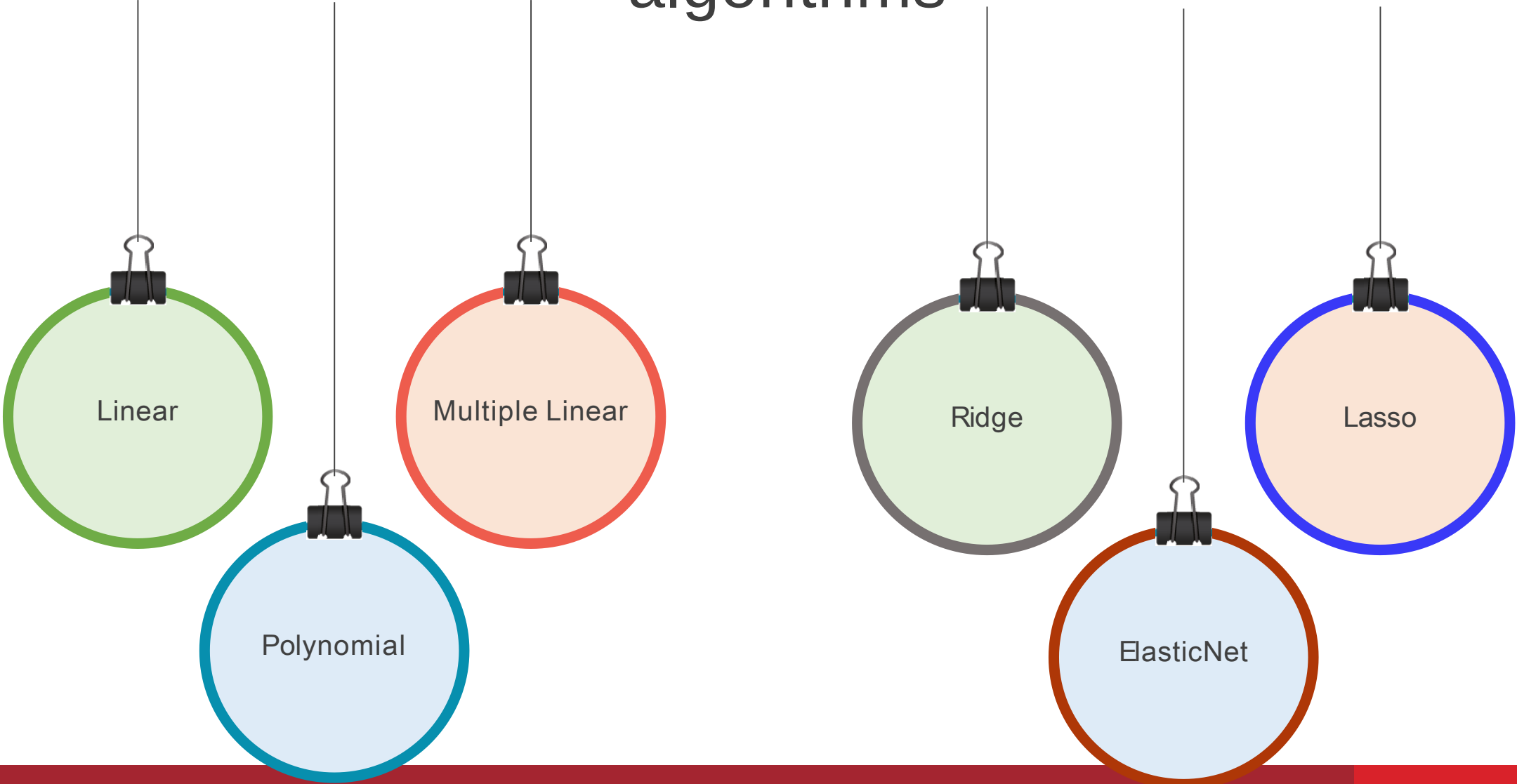
Dependent Variable: The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called target variable.

Independent Variable: The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a predictor.

Outliers: Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.

Underfitting and Overfitting

Types of regression algorithms



Linear

Multiple Linear

Ridge

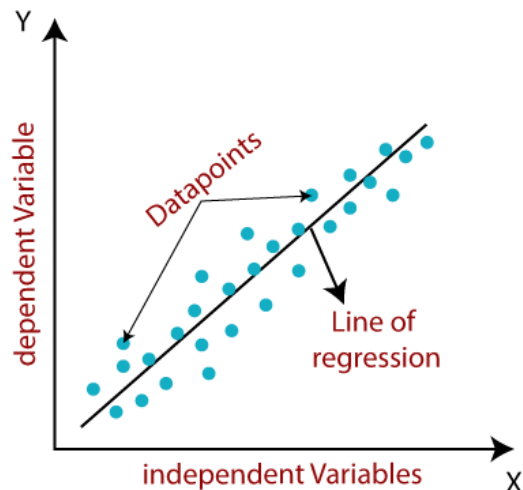
Lasso

Polynomial

ElasticNet

Types of Regression Algorithms

Linear Regression is a statistical model used to predict the relationship between independent and dependent variables denoted by x and y respectively



Examine 2 factors

1

How closely are x and y related ?

Linear regression gives a number between -1 and 1 indicating the strength of correlation between the two variables

- 1 : no correlation
- 2 : positively correlated
- 1 : negatively correlated

2

Prediction

When the relationship between x and y is known, use this to predict future values of y for a value of x

This is done by fitting a regression line and represented by a linear equation:

$$y = m * x + b$$

Types of Linear Regression

Simple Linear Regression:

If a single independent variable is used to predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Simple Linear Regression.

Multiple Linear regression:

If more than one independent variable is used to predict the value of a numerical dependent variable, then such a Linear Regression algorithm is called Multiple Linear Regression.

Simple Linear Regression

- **Model the relationship between the two variables.** Such as the relationship between Income and expenditure, experience and Salary, etc.
- **Forecasting new observations.** Such as Weather forecasting according to temperature, Revenue of a company according to the investments in a year, etc.

Error, or residual.

Difference of actual and predicted value of Y given by,

$$\text{Error}, E_i = Y_i - Y_P^i$$

$$Y_P^i = m_P \cdot X_i + C_P$$

$$\text{Error}, E_i = Y_i - m_P \cdot X_i - C_P$$

Error, or residual.

According to Least Square Approximation, the sum of squared error can be written as,

$$S.S.E = \sum_{i=1}^N E_i^2$$

The best fit line can be obtained by reducing this error to the least value possible which can be done by setting the gradient to zero.

$$\frac{d (\sum_{i=1}^N E_i^2)}{d C_P} = 0 \quad \dots (1)$$

$$\frac{d (\sum_{i=1}^N E_i^2)}{d m_P} = 0 \quad \dots (2)$$

Error, or residual.

$$\frac{d (\sum_{i=1}^N (Y_i - m_P \cdot X_i - C_P)^2)}{d C_P} = 0$$

$$\sum_{i=1}^N 2(Y_i - m_P \cdot X_i - C_P) \cdot (-1) = 0$$

$$\sum_{i=1}^N Y_i - \sum_{i=1}^N m_P \cdot X_i - \sum_{i=1}^N C_P = 0$$

$$\bar{Y} - m_P \cdot \bar{X} - C_P = 0 \quad \dots (3)$$

Mean of dependent variable, $\bar{Y} = \frac{\sum_{i=1}^N Y_i}{N}$

Mean of independent variable, $\bar{X} = \frac{\sum_{i=1}^N X_i}{N}$

Error, or residual.

$$\frac{d (\sum_{i=1}^N (Y_i - m_P \cdot X_i - C_P)^2)}{d m_P} = 0$$

$$\sum_{i=1}^N 2(Y_i - m_P \cdot X_i - C_P) \cdot (-X_i) = 0$$

$$\sum_{i=1}^N X_i Y_i - m_P \sum_{i=1}^N X_i^2 - C_P \cdot \sum_{i=1}^N X_i = 0 \quad \dots (4)$$

Solving equation 3 and 4 for slope and intercept of best fit predicted line,

$$C_P = \frac{(N \cdot \sum_{i=1}^N X_i \cdot Y_i) - (\sum_{i=1}^N X_i \cdot \sum_{i=1}^N Y_i)}{N \cdot \sum_{i=1}^N X_i^2 - (\sum_{i=1}^N X_i)^2}$$

$$m_P = \frac{(\sum_{i=1}^N Y_i) - C_P (\sum_{i=1}^N X_i)}{N}$$

$$= \bar{Y} - C_P \cdot \bar{X}$$

Example - 1

Find the best-fit line to predict weight from height. Use Simple Linear Regression

x = independent variable (Height)

y = dependent variable (Weight)

m = slope

b = intercept

Goal: Find values of m and b that best fit the data.

Sr. No.	Height (cm)	Weight (kg)
1	151	63
2	174	81
3	138	56
4	186	91
5	128	47
6	136	57
7	179	76
8	163	72
9	152	62
10	131	48

Example - 1

- Step 1 (Mean)

$$\bar{x} = \Sigma x / n$$

$$\bar{y} = \Sigma y / n$$

- Step 2 (Slope Formula)

$$m = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$$

- Step 3 (Intercept)

$$b = \bar{y} - m \bar{x}$$

$$b = 65.3 - (0.675 \times 153.8)$$

$$b \approx -38.46$$

Sr. No.	Height (cm)	Weight (kg)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x}) \cdot (y - \bar{y})$	$(x - \bar{x})^2$
1	151	63	-2.8	-2.3	6.44	7.84
2	174	81	20.2	15.7	317.14	408.04
3	138	56	-15.8	-9.3	146.94	249.64
4	186	91	32.2	25.7	827.54	1036.84
5	128	47	-25.8	-18.3	472.14	665.64
6	136	57	-17.8	-8.3	147.74	316.84
7	179	76	25.2	10.7	269.64	635.04
8	163	72	9.2	6.7	61.64	84.64
9	152	62	-1.8	-3.3	5.94	3.24
10	131	48	-22.8	-17.3	394.44	519.84
Sum	1538	653			2649.6	3927.6
Mean	153.8	65.3				

Prediction Example 1

For height = 160 cm:

$$y = m * x + b$$

$$y = 0.675(160) - 38.46$$

$$y \approx 69.5 \text{ kg}$$

Example 2

Suppose we are given a dataset. Given is a Salary vs Experience dataset of a company and the task is to predict the salary of an employee based on his / her work experience. Prepare the equation for linear regression model and predict the salary of an employee if he has an experience of 4 years.

Experience (X)	Salary (y) (in lakhs)
2	3
6	10
5	4
7	13

Example 2

Hence, using linear regression we can predict the salary as shown below:

x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
2	3	-3	-4.5	9	13.5
6	10	1	2.5	1	2.5
5	4	0	-3.5	0	0
7	13	2	5.5	4	11
$\bar{x} = 5$	$\bar{y} = 7.5$			$\Sigma = 14$	$\Sigma = 27$

$$\beta_1 = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\Sigma(x_i - \bar{x})^2}$$

$$= \frac{27}{14} = 1.928$$

$$\beta_1 = 1.928$$

and,

$$\begin{aligned}\beta_0 &= \bar{y} - \beta_1 \bar{x} \\ &= 7.5 - (1.928 \times 5) \\ &= -2.14\end{aligned}$$

$$\beta_0 = -2.14$$

Example 2

Therefore, the equation of the regression line is given as:

$$y = -2.14 + 1.928x$$

Now,

Suppose if an employee has an experience of 4 years, then his/her salary will be predicted as:

$$y = -2.14 + (1.928 \times 4) = 5.572$$

Practice Example

Given is a Size of a flat in Sq.ft. vs Flat Prize dataset of a real estate company and the task is to predict prize of a flat based on the size. Prepare the equation for linear regression model and predict the prize of a flat of size 700 sq. ft.

Size(sq.ft)	Prize(Lakh)
450	50
500	55
575	65
600	75
850	97

Pearson's Coefficient

- The first person to give a mathematical formula for the measurement of the degree of relationship between two variables in 1890 was Karl Pearson.
- The correlation coefficient is a measure of the strength and direction of the correlation between the independent variable and the dependent variable .
- The formula for r is shown

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \sqrt{n\sum y^2 - (\sum y)^2}}$$

Pearson's Coefficient (contd)

Step	Representation in Symbols
1. Calculate the sum of the x -values.	Σx
2. Calculate the sum of the y -values.	Σy
3. Multiply each x -value by the corresponding y -value and calculate the sum of these xy products.	Σxy
4. Square each x -value and then calculate the sum of these squared values.	Σx^2
5. Square each y -value and then calculate the sum of these squared values.	Σy^2
6. Determine the value of n , which is the number of data pairs.	n
7. Use these results to then substitute into the formula for the correlation coefficient.	$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2} \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$

Interpretation of correlation coefficient

Once the value of r is calculated, this measurement provides two indicators for the correlation:

- 1.the strength of the correlation based on the *value* of r
- 2.the direction of the correlation based on the *sign* of r

The value of r gives us information

- The value of r is always between -1 and $+1$: $-1 \leq r \leq 1$.
- The size of the correlation r indicates the strength of the linear relationship between the two variables. Values of r close to -1 or $+1$ to indicate a stronger linear relationship.
- If $r=0$, there is no linear relationship between the two variables (*no linear correlation*).
- If $r=1$, there is perfect positive correlation. If $r=-1$, there is perfect negative correlation. In both of these cases, all the original data points lie on a straight line.

The sign of r gives us information

- A positive value of r means that when x increases, y tends to increase, and when x decreases, y tends to decrease (*positive correlation*).
- A negative value of r means that when x increases, y tends to decrease, and when x decreases, y tends to increase (*negative correlation*).

Example 1 with Pearson's Coefficient

Compute the correlation coefficient for the given data to find the relation between height and weight.

Use formula

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \sqrt{n\sum y^2 - (\sum y)^2}}$$

Sr. No.	Height (cm)	Weight (kg)
1	151	63
2	174	81
3	138	56
4	186	91
5	128	47
6	136	57
7	179	76
8	163	72
9	152	62
10	131	48

Example 1 with Pearson's Coefficient

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$r = \frac{(10 \times 103081) - (1538 \times 653)}{\sqrt{[(10 \times 240272) - 1538^2]} \sqrt{[(10 \times 44513) - 653^2]}}$$

$$r = 0.9771$$

Sr. No.	Height (cm)	Weight (kg)	xy	x ²	y ²
1	151	63	9513	22801	3969
2	174	81	14094	30276	6561
3	138	56	7728	19044	3136
4	186	91	16926	34596	8281
5	128	47	6016	16384	2209
6	136	57	7752	18496	3249
7	179	76	13604	32041	5776
8	163	72	11736	26569	5184
9	152	62	9424	23104	3844
10	131	48	6288	17161	2304
Sum	1538	653	103081	240472	44513
Mean	153.8	65.3			

Multiple Linear Regression

- In the Simple Linear Regression a single Independent/Predictor(X) variable is used to model the response variable (Y). But there may be various cases in which the response variable is affected by more than one predictor variable; for such cases, the Multiple Linear Regression algorithm is used.
- Moreover, Multiple Linear Regression is an extension of Simple Linear regression as it takes more than one predictor variable to predict the response variable. We can define it as:
- *“Multiple Linear Regression is one of the important regression algorithms which models the linear relationship between a single dependent continuous variable and more than one independent variable.”*
- e.g. Prediction of CO₂ emission based on engine size and number of cylinders in a car

Multiple Linear Regression

Linear
Regression

Multiple
Linear
Regression

Ridge
Regression

Lasso
Regression

ElasticNet
Regression

Multiple linear regression is a statistical technique used to predict the outcome of a response variable through several explanatory variables and model the relationships between them.

Equation for MLR

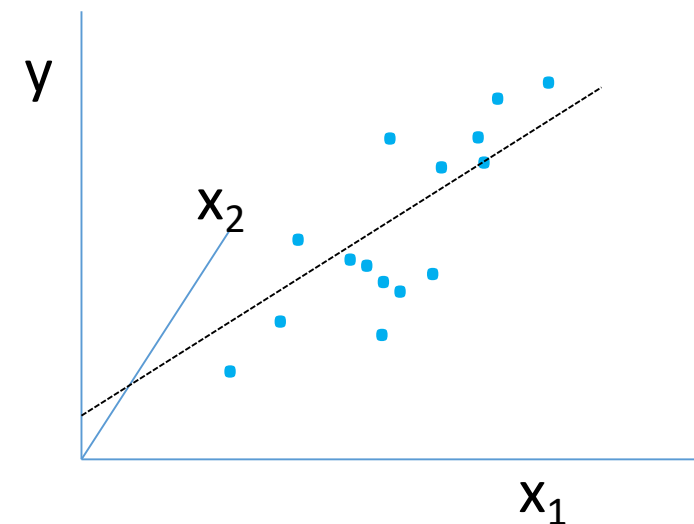
$$Y = m_1 * x_1 + m_2 * x_2 + m_3 * x_3 + \dots + m_n * x_n + c$$

Dependent Variable

$m_1, m_2, m_3 \dots m_n$

Slopes

Coefficient



Multiple Linear Regression

Introduce the matrix

$$Y = X\beta$$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad \beta = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \end{bmatrix}$$

Where

- $X \rightarrow$ input matrix
- $\beta \rightarrow$ parameter vector
- $Y \rightarrow$ output vector

Coefficient Estimation

$$\beta = (X^T X)^{-1} X^T Y$$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \quad \beta = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \end{bmatrix}$$

Where

- $X \rightarrow$ input matrix
- $\beta \rightarrow$ parameter vector
- $Y \rightarrow$ output vector

Example 3

Apply multiple linear regression.

Sr. No.	Height	Age	Weight
1	151	20	63
2	174	22	81
3	138	19	56
4	186	25	91
5	128	18	47

$$y = b_0 + b_1x_1 + b_2x_2$$

Example 3 (contd)

Form the Matrix X and Y

$$X = \begin{bmatrix} 1 & 151 & 20 \\ 1 & 174 & 22 \\ 1 & 138 & 19 \\ 1 & 186 & 25 \\ 1 & 128 & 18 \end{bmatrix}$$

$$Y = \begin{bmatrix} 63 \\ 81 \\ 56 \\ 91 \\ 47 \end{bmatrix}$$

$$\beta = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 151 & 174 & 138 & 186 & 128 \\ 20 & 22 & 19 & 25 & 18 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 777 & 104 \\ 777 & 123101 & 16424 \\ 104 & 16424 & 2194 \end{bmatrix}$$

$$(X^T X)^{-1} \approx \begin{bmatrix} 18.2211 & 0.1821 & -2.2274 \\ 0.1821 & 0.00836 & -0.07118 \\ -2.2274 & -0.07118 & 0.6389 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 338 \\ 55699 \\ 7217 \end{bmatrix}$$

Example 3 (contd)

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = (X^T X)^{-1} (X^T Y)$$

$$\beta \approx \begin{bmatrix} 18.2211 & 0.1821 & -2.2274 \\ 0.1821 & 0.00836 & -0.07118 \\ -2.2274 & -0.07118 & 0.6389 \end{bmatrix} \begin{bmatrix} 338 \\ 55699 \\ 7217 \end{bmatrix}$$

$$\beta = \begin{bmatrix} -51.16 \\ 0.64 \\ 0.91 \end{bmatrix}$$

$$y = -51.16 + 0.64(\text{Height}) + 0.91(\text{Age})$$

Error in Regression – What are we actually measuring?

In regression, prediction is never perfect.
Every model makes mistake~

$$\text{Error} = y - \hat{y}$$

“How far is my model from reality?”

Mean Squared Error (MSE) – Optimization Perspective

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

- **Pros:** MSE penalizes large errors more, which can help when it's important to avoid large mistakes in predictions.
- **Cons:** It's sensitive to outliers, and it may not be easy to interpret because it's in squared units of the target variable.

RMSE

These functions are called the loss function or the cost function, and the value has to be minimized.

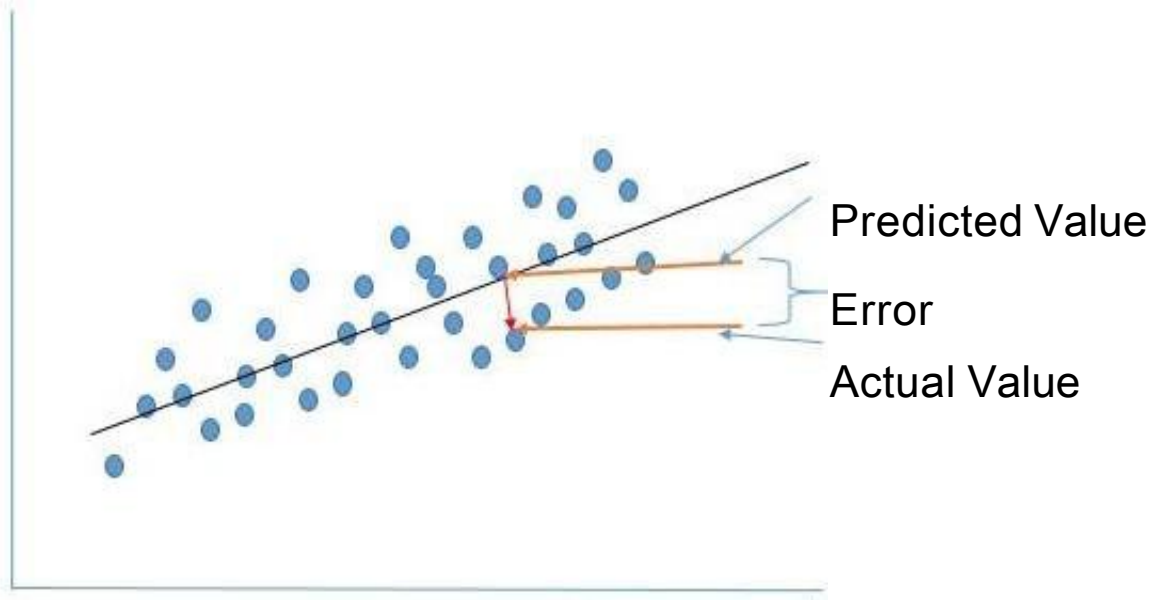
$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE) – Robustness Perspective

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

- **Pros:** MAE is easier to interpret because it's in the same unit as the target variable and is less sensitive to outliers.
- **Cons:** Unlike MSE and RMSE, MAE doesn't penalize larger errors as heavily, which could be a disadvantage in some cases where large errors are particularly undesirable.

R² Score



$$\text{R-square} = 1 - \frac{\sum(Y_{\text{actual}} - Y_{\text{predicted}})^2}{\sum(Y_{\text{actual}} - Y_{\text{mean}})^2}$$

R-square is the most common metric to judge the performance of regression models.

R² lies between 0 -100%

Example: Performing linear regression on sq. Area (x) and Price (y) returns **R-square** value as 16. This means you have 16% information to make an accurate prediction about the price.

Adjusted R-Squared

The disadvantage with R-squared is that it assumes every independent variable in the model explains variations in the dependent variable.

Use adjusted R-squared when working on a multiple linear regression problem.

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

where R^2 is R-squared value

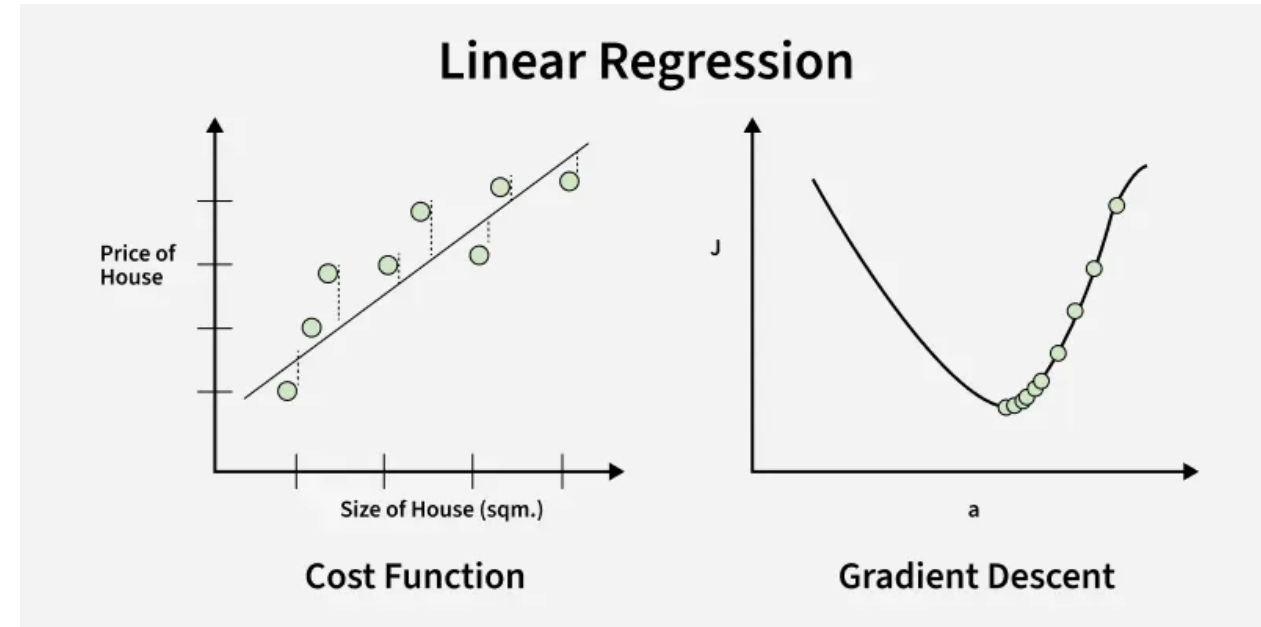
p is number of predictor variables

N is number datapoints

Metrics	What?	Why?	When to Use?
<u>MSE</u>	Measures average squared difference between estimated and actual values.	Emphasizes larger errors.	When large errors are more critical.
<u>RMSE</u>	Square root of MSE, in same units as response variable.	Easier interpretation of errors.	When error scale should match target scale.
<u>MAE</u>	Average absolute difference between estimated and actual values.	Less sensitive to outliers.	With many outliers or non-normal residuals.
<u>MAPE</u>	Percentage error between estimated and actual values.	Easy interpretation as a percentage.	For forecasting and percentage-based error analysis.
<u>R-Squared</u>	Proportion of variance explained by the model.	Indicates model's explanatory power.	To evaluate linear regression models' fit.
<u>Adjusted R-squared</u>	Statistical measure that modifies the R-Squared value to account for the number of predictors	Unlike R-squared, it penalizes the model for including irrelevant predictors	Useful in multiple regression scenarios where you have several independent variables

Gradient Descent

- Gradient descent is another algorithm used to reduce the loss function.
- It is an optimization algorithm that tweaks its parameters (coefficients) iteratively to minimize a given cost function to its minimum.
- Model stops learning when the gradient (slope) is zero



Gradient Descent - Algorithm

1. Initialize Parameters

Start with random initial values for the slope (m) and intercept (b).

2. Define the Cost Function

Measure the error between predicted and actual values using the **Mean Squared Error (MSE)**

$$J(m, b) = \frac{1}{n} \sum_{i=1}^n (y_i - (mx_i + b))^2$$

where:

n = number of data points

y_i = actual value

mx_i+b = predicted value

Gradient Descent - Algorithm

3. Compute the Gradients

Calculate the partial derivatives of the cost function with respect to m and b :

For slope (m): $\frac{\partial J}{\partial m} = -\frac{2}{n} \sum_{i=1}^n x_i (y_i - (mx_i + b))$ For intercept (b): $\frac{\partial J}{\partial b} = -\frac{2}{n} \sum_{i=1}^n (y_i - (mx_i + b))$

4. Update Parameters

Adjust m and b using the learning rate (α):

$$m = m - \alpha \cdot \frac{\partial J}{\partial m}$$

$$b = b - \alpha \cdot \frac{\partial J}{\partial b}$$

Here, α controls the step size of each update.

Example – Gradient Descent

Multiple Linear Regression with Gradient Descent

Applicant	Income (Lakhs) (X_1)	Credit Score (X_2)	Loan Approved (Y)
1	5.0	645	19.2
2	6.5	720	27.1
3	7.0	766	31.5
4	8.5	810	38.4
5	9.0	880	42.8

Predict Loan Approved (Y) using: Annual Income (X_1) and Credit Score (X_2)

$$\text{Model: } \hat{y} = m_1x_1 + m_2x_2 + b$$

Initialization

$m_1 = 0, m_2 = 0, b = 0$
Learning Rate $\alpha = 0.0000001$

Applicant	Income (Lakhs) (X_1)	Credit Score (X_2)	Loan Approved (Y)
1	5.0	645	19.2
2	6.5	720	27.1
3	7.0	766	31.5
4	8.5	810	38.4
5	9.0	880	42.8

Cost Function

$$J = (1/2n) \sum (\hat{y} - y)^2$$

Initially: $\hat{y}_i = 0$

Applicant	Income (Lakhs) (X_1)	Credit Score (X_2)	Loan Approved (Y)
1	5.0	645	19.2
2	6.5	720	27.1
3	7.0	766	31.5
4	8.5	810	38.4
5	9.0	880	42.8

$$J = \frac{1}{10} (19.2^2 + 27.1^2 + 31.5^2 + 38.4^2 + 42.8^2)$$

$$J = \frac{1}{10} (368.64 + 734.41 + 992.25 + 1474.56 + 1831.84)$$

$$J = \frac{1}{10} (5401.7) = 540.17$$

Gradient Formulas

$$\begin{aligned}\frac{\partial J}{\partial m_1} &= (1/n) \sum (\hat{y} - y)x_1 \\ \frac{\partial J}{\partial m_2} &= (1/n) \sum (\hat{y} - y)x_2 \\ \frac{\partial J}{\partial b} &= (1/n) \sum (\hat{y} - y)\end{aligned}$$

Since $\hat{y}_i = 0$,
 $(\sum (\hat{y}_i - y_i) = -y_i$

$$\begin{aligned}\frac{\partial J}{\partial b} &= \frac{1}{5}(-19.2 - 27.1 - 31.5 - 38.4 - 42.8) \\ &= \frac{-159}{5} = -31.8\end{aligned}$$

$$\begin{aligned}\frac{\partial J}{\partial m_1} &= \frac{1}{5}[-(19.2 \times 5.0 + 27.1 \times 6.5 + 31.5 \times 7.0 + 38.4 \times 8.5 + 42.8 \times 9.0)] \\ &= \frac{1}{5}[-(96 + 176.15 + 220.5 + 326.4 + 385.2)] \\ &= \frac{-1204.25}{5} \\ &= -240.85\end{aligned}$$

$$\begin{aligned}\frac{\partial J}{\partial m_2} &= \frac{1}{5}[-(19.2 \times 645 + 27.1 \times 720 + 31.5 \times 765 + 38.4 \times 810 + 42.8 \times 880)] \\ &= \frac{1}{5}[-(12384 + 19512 + 24097.5 + 31104 + 37664)] \\ &= \frac{-124761.5}{5} \\ &= -24952.3\end{aligned}$$

Gradient Values

$$\partial J / \partial m_1 = -240.85$$

$$\partial J / \partial m_2 = -24952.3$$

$$\partial J / \partial b = -31.8$$

Now update the parameter

$$m_1 = m_1 - \alpha \frac{\partial J}{\partial m_1}$$

$$m_2 = m_2 - \alpha \frac{\partial J}{\partial m_2}$$

$$b = b - \alpha \frac{\partial J}{\partial b}$$

$$m_1 = 0 - (0.000001)(-240.85) =$$

$$m_2 = 0 - (0.000001)(-24952.3) =$$

$$b = 0 - (0.000001)(-31.8) =$$

Parameter Update

$$m_1 = 0.00024085$$

$$m_2 = 0.0249523$$

$$b = 0.0000318$$

New regression equation after 1st iteration

$$\hat{y} = 0.00024085x_1 + 0.0249523x_2 + 0.0000318$$



SOMAIYA
VIDYAVIHAR UNIVERSITY

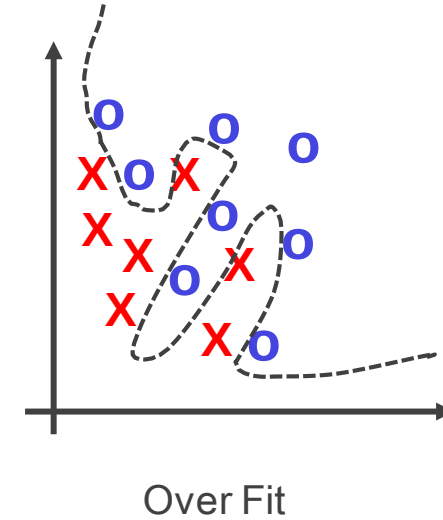
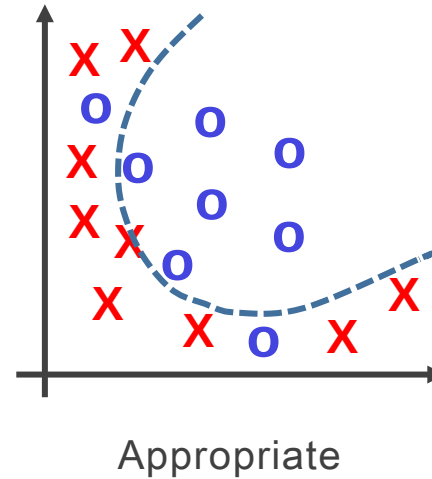
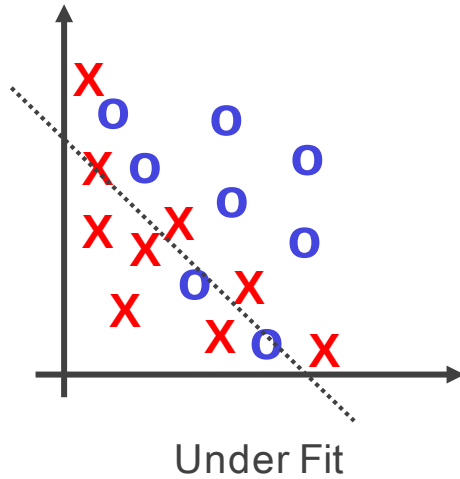
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Updated Equation



$$\hat{y} = 0.00024085x_1 + 0.0249523x_2 + 0.0000318$$

Challenges in Prediction



If the model learning is poor, you have an **underfitted** situation

The algorithm will not work well on test data
Retraining may be needed to find a better fit

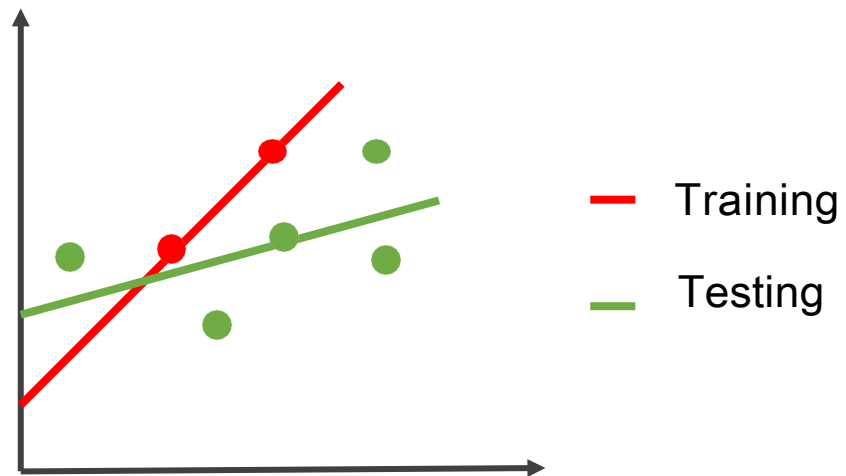
Overfitting happens when model accuracy for training data is good, but model does not generalize well to the overall population

Algorithm is not able to give good predictions for the new data

Regularization

Regularization solves overfitting to the training data.

Used to restrict the parameters values that are estimated in the model



$$L = \sum (\hat{Y}_i - Y_i)^2 + \lambda \sum \beta^2$$

This loss function includes 2 elements.

- 1) the sum of square distances between predicted and actual value
- 2) the second element is the regularization term

Types of Regression (Contd.)

Linear
Regression

Ridge Regression (L2) is used when there is a problem of multicollinearity.

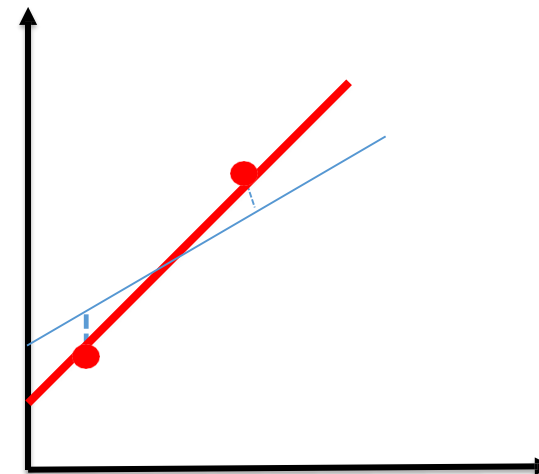
By adding a degree of bias to the regression estimates, ridge regression reduces the standard errors.

Multiple
Linear
Regression

The main idea is to find a new line that has some bias with respect to the training data

In return for that small amount of bias, a significant drop in variance is achieved

Ridge
Regression



Lasso
Regression

Minimization objective = $SSE + \lambda * (\text{sum of the square of coefficients})$

ElasticNet
Regression

λ controls the strength of the penalty term

Types of Regression (Contd.)

Linear
Regression

Lasso Regression (L1) is similar to ridge, but it also performs feature selection.

It will set the coefficient value for features that do not help in decision making very low, potentially zero.

Minimization objective = $SSE + \lambda * (\text{sum of absolute coefficient values})$

Ridge
Regression

Lasso regression tends to exclude variables that are not required from the equation, whereas ridge tends to do better when all variables are present.

Lasso
Regression

ElasticNet
Regression

Types of Regression (Contd.)

ElasticNet regression combines
the strength of **lasso** and **ridge regression**



$$\begin{aligned} &\text{the sum of the squared residuals} \\ &+ \\ &\lambda_1 \times |\text{variable}_1| + \dots + |\text{variable}_x| \quad + \quad \lambda_2 \times \text{variable}_1^2 + \dots + \text{variable}_x^2 \end{aligned}$$



Lasso penalty



Ridge penalty

If you are not sure whether to use lasso or ridge, use ElasticNet

Linear
Regression

Multiple
Linear
Regression

Ridge
Regression

Lasso
Regression

ElasticNet
Regression

Example – Apply Ridge and Lasso

- Step 1 (Mean)

$$\bar{x} = \Sigma x / n$$

$$\bar{y} = \Sigma y / n$$

- Step 2 (Slope Formula)

$$m = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$$

- Step 3 (Intercept)

$$b = \bar{y} - m \bar{x}$$

$$b = 65.3 - (0.675 \times 153.8)$$

$$b \approx -38.46$$

Sr. No.	Height (cm)	Weight (kg)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x}) \cdot (y - \bar{y})$	$(x - \bar{x})^2$
1	151	63	-2.8	-2.3	6.44	7.84
2	174	81	20.2	15.7	317.14	408.04
3	138	56	-15.8	-9.3	146.94	249.64
4	186	91	32.2	25.7	827.54	1036.84
5	128	47	-25.8	-18.3	472.14	665.64
6	136	57	-17.8	-8.3	147.74	316.84
7	179	76	25.2	10.7	269.64	635.04
8	163	72	9.2	6.7	61.64	84.64
9	152	62	-1.8	-3.3	5.94	3.24
10	131	48	-22.8	-17.3	394.44	519.84
Sum	1538	653			2649.6	3927.6
Mean	153.8	65.3				

Summary

 Concept	 Formula / Explanation
◆ Simple Linear Regression	$Y = \beta_0 + \beta_1 X$
◆ Multiple Linear Regression	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$
◆ Coefficient (β_1) Formula	$\beta_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$
◆ Intercept (β_0) Formula	$\beta_0 = \bar{Y} - \beta_1 \bar{X}$
◆ Matrix Formula for Multiple Regression	$\beta = (X^T X)^{-1} X^T Y$
◆ Coefficient Interpretation	 Positive ($\beta > 0$) = Direct relationship,  Negative ($\beta < 0$) = Inverse relationship